

Tim J. Schoonbeek

PHD CANDIDATE · PERCEPTION · MULTIMODAL MACHINE LEARNING

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Summary

PhD candidate at ASML and TU/e (graduating November 25). I build scalable perception for industrial augmented reality applications, ranging from multimodal video models for step recognition to assembly object detection and error localization based on 3D CAD models. I care about bridging the gaps between AI research and real-world applications, as demonstrated through publications in computer vision (WACV), robotics (ICRA), and autonomous driving (IV).

TECHNICAL SKILLS

Deep learning: CV, 3D perception, ML, augmented reality, HCI, video understanding, transformers, convnets, contrastive learning.
Languages & tools: Python, PyTorch, NumPy, OpenCV, Unity, Pandas, Azure, Git, (C++).

Experience

Microsoft Research

Redmond, USA

RESEARCH INTERN: INTERACTIVE MULTIMODAL AI SYSTEMS

June 2025 - Sept. 2025

- Developing real-time multimodal systems with LLM-powered perception to create benchmarks for physically situated intelligence.
- Building models of social cues (turn-taking, engagement, attention) from large datasets while collaborating with cross-disciplinary teams.

ASML Research

Eindhoven, the Netherlands

VISITING RESEARCHER: PHD ON ADVANCED AUGMENTED REALITY SOLUTIONS FOR AI-BASED SERVICITIZATION

Sept. 2021 - Nov. 2025

- PhD project on automatic extraction of relevant and meaningful information from procedural actions within an industrial setting.
- Filed patents, published peer-reviewed articles, and contributed to the ASML Technology Conference, the world's largest developer event.
- Built egocentric industrial dataset & evals (IndustReal); released tasks and metrics; benchmarked with several backbones (5k+ downloads).
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Honda Research Institute Europe GmbH

Frankfurt, Germany

MASTER INTERNSHIP: INTERACTION-AWARE TRAJECTORY PREDICTION USING GRAPH NEURAL NETWORKS

Mar. 2020 - July 2020

- Researched spatio-temporal graph neural networks for trajectory prediction of surrounding vehicles in highway driving scenarios.

Automotive Technology InMotion (Student Team)

Helmond, the Netherlands

RACE ENGINEER - PART-TIME

Sept. 2018 - Sept. 2019

- Performed and optimized drive cycle simulations for an electric endurance racing car.

Education

Eindhoven University of Technology

Eindhoven, Netherlands

PHD ON ADVANCED AUGMENTED REALITY SOLUTIONS FOR AI-BASED SERVICITIZATION (ELECTRICAL ENGINEERING FACULTY)

Sept. 2021 - Nov. 2025

- Doctoral project funded by ASML on industrial procedure understanding using computer vision and machine learning algorithms.
- Project on the intersection of computer vision, human-computer interaction, and automation.
- Focus on scalable, multimodal computer-vision systems for procedure understanding and creating relevant industrial benchmarks.

University of Cambridge

Cambridge, United Kingdom

VISITING RESEARCHER

Oct. 2024 - April 2025

- Conducted research on enhancing human-robot interaction, focusing on AI systems to assist industrial operators in procedure execution.
- Designed perception algorithms for error localization for a real-time deployment on AR glasses; conducted user studies (N=24).
- Reached the finals (top 2.5%) in the OxBridge AI Startup Challenge, a competition between students from Cambridge and Oxford.

International Summer School

Matera, Italy

EXTENDED REALITY AND ARTIFICIAL INTELLIGENCE

July 2023

- Pro-actively took up the role as a link between the programmers and designers in a multidisciplinary project, besides my technical roles.
- Won the *best project award* for our work on an XR app for engagement with cultural heritage.

Eindhoven University of Technology

Eindhoven, Netherlands

M.Sc. IN AUTOMOTIVE ENGINEERING (SPECIALIZATION: MOBILE PERCEPTION SYSTEMS)

Feb. 2019 - May 2021

- Earned the *cum laude* honours distinction for a 4.0 GPA (8.5/10).
- Master thesis "Learning to Predict Collision Risk from Simulated Optical Flow" awarded a 9.0/10.0. Conducted research on sim-to-real domain generalization by design, rather than through explicit domain adaptation techniques.
- Published paper, singled out among the top 5% of submissions for *oral presentation* at the 2022 IEEE Intelligent Vehicles Symposium.

Eindhoven University of Technology

Eindhoven, Netherlands

B.Sc. IN ELECTRICAL ENGINEERING (AUTOMOTIVE TRACK)

Sept. 2015 - Feb. 2019

- Bachelor thesis on depth estimation from disparity and segmented images awarded a 8.5/10.

Selected publications

WACV 2024	Tim J. Schoonbeek <i>et al.</i> , IndustReal: A Dataset for Procedure Step Recognition Handling Execution Errors in Egocentric Videos in an Industrial-Like Setting
CVIU (J.)	Tim J. Schoonbeek <i>et al.</i> , Learning to Recognize Correctly Completed Procedure Steps in Egocentric Assembly Videos through Spatio-Temporal Modeling
ICRA & RA-L	Tim J. Schoonbeek <i>et al.</i> , Supervised Representation Learning towards Generalizable Assembly State Recognition
ECCV-W 2024	Tim J. Schoonbeek and Dan Lehman <i>et al.</i> , Find the Assembly Mistakes: Error Segmentation for Industrial Applications
IEEE IV 2022	Tim J. Schoonbeek <i>et al.</i> , Learning to Predict Collision Risk from Simulated Video Data

Industry Contributions

EP Patent (<i>Publ.</i>)	Tim J. Schoonbeek <i>et al.</i> , Defect Identification and Segmentation without Labeled Data and Image Quality Enhancement
WO Patent (<i>Publ.</i>)	Tim J. Schoonbeek <i>et al.</i> , Contrastive Deep Learning for Scanning Electron Microscope Defect Inspection
ASML TC 2024	Tim J. Schoonbeek <i>et al.</i> , Verifying Procedure States of Assemblies using Contrastive Learning and Synthetic Data
ASML TC 2024	Thomas Van de Moosdijk <i>et al.</i> , Superimposed Work Instructions using Augmented Reality
ASML TC 2023	Tim J. Schoonbeek <i>et al.</i> , Automated Recognition of Procedural Actions for Real-Time Understanding of Service Tasks